Exponential growth and decay

1

a

i Exponential growth

ii 0.6

b

i Exponential decay

ii 0.8

2 233

3 $f\left(t\right) = 26030 \times \left(1.011\right)^{t}$

4 $P(t) = 10\,637\left(1.0040\right)^{t}$

5

a $y = 750\left(1.03\right)^{t}$

b 869

6

a $y = 2900\left(1.09\right)^{v}$

b $\$5778.43$

7

a $y = 322\,000\left(0.95\right)^{m}$

b $\$249\,157.46$

8

a $y = 2310\left(0.974\right)^{n}$

b 1328

9

a $y = 200\left(3\right)^{n}$

b $\$48\,600$

10

a $y = 21\left(0.5\right)^{n}$

b 0.66 m

11

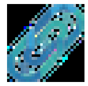
a $A = 300\left(0.5\right)^{\frac{t}{20}}$

b 0.59 grams

12

a Yes **b** No **c** No **d** Yes

13 $r = 3.53\%$



14 631 bacteria

15

a 0.01

b $A = 38\,200 \times 0.99^n$

c $\$35\,964.54$

d Motorbike

16 $41\,199$

17 6591

18 262

19

a

Quarter(N)	1	2	3	4	5	6	7
Online Customers (C)	480	2880	17280	103 680	622 080	3 732 480	22 394 880

b $C = 480 \times 6^{N-1}$

c $29\,023\,764\,480$ sales

d To not assume that growth will change in exactly the same way in the coming quarters and try a different model.